

Karnataka State Board

Science Unit Test - Photosynthesis

Subject: Science

Grade: 8

Maximum Marks: 9

All questions are compulsory.

Marks for each question are shown against it.

1. Compare the light-dependent reactions and the light-independent (Calvin) reactions of photosynthesis, highlighting three key differences. **[3]**
2. Describe the overall process of photosynthesis in green plants, mentioning the main stages and their locations within the chloroplast. **[3]**
3. Explain the role of stomata in the photosynthetic process of a leaf. **[3]**

Answer Key

1. 1. Light-dependent reactions occur in the thylakoid membranes and require light to produce ATP and NADPH. 2. Light-independent reactions (Calvin cycle) take place in the stroma and use ATP and NADPH to fix CO₂ into glucose. 3. Oxygen is released as a by-product of the light-dependent reactions, whereas the Calvin cycle does not produce O₂.

In a labeled diagram of a chloroplast, the thylakoid stacks (granum) are shown where photolysis of water and electron transport generate ATP and NADPH. These energy carriers move to the stroma, where the Calvin cycle incorporates carbon dioxide into carbohydrate molecules, completing the synthesis of glucose without releasing oxygen.

2. 1. Light energy is absorbed by chlorophyll pigments in the thylakoid membranes of the chloroplast. 2. The absorbed energy is used to produce ATP and NADPH in the light-dependent reactions. 3. ATP and NADPH drive the Calvin cycle in the stroma, fixing CO₂ into glucose.

The chloroplast is divided into thylakoids where light-dependent reactions occur, generating the energy carriers ATP and NADPH. These carriers then move to the stroma, where the light-independent Calvin cycle incorporates carbon dioxide into glucose molecules, completing photosynthesis as shown in a labeled diagram of a chloroplast.

3. 1. Stomata open to allow CO₂ from the atmosphere to enter the leaf for carbon fixation. 2. They provide an exit route for the O₂ produced during photosynthesis. 3. Stomatal opening also facilitates transpiration, helping to maintain leaf temperature and water flow.

By regulating gas exchange, stomata ensure that sufficient carbon dioxide reaches the mesophyll cells where the Calvin cycle occurs, while the released oxygen diffuses out. Transpiration through stomata creates a pull that draws water up from the roots, supporting the photosynthetic reactions.